SEPARATED-CATEGORY COLIMITS BECOME HONEST COLIMITS UNDER FIVE HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 556-574 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 576-610, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_spaces-pushouts.json (pb=1936, stmt 556-574, proof 576-610).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GHL: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 450–469. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $\mathcal{I} \to (Sch/S)_{fppf}$, $i \mapsto X_i$ be a diagram of algebraic spaces over B. Assume that*

1. *each X_i is separated over B,*
2. *$X = \operatorname{colim} X_i$ exists in the category of algebraic spaces separated over B,*
3. *$\coprod X_i \to X$ is surjective,*
4. *if $U \to X$ is an étale separated morphism of algebraic spaces and $U_i = X_i \times_X U$, then $U = \operatorname{colim} U_i$ in the category of algebraic spaces separated over B, and*
5. *every object $(U_i \to X_i)$ of $\lim X_{i,spaces,étale}$ with $U_i \to X_i$ separated is of the form $U_i = X_i \times_X U$ for some étale separated morphism of algebraic spaces $U \to X$.*

Then $X = \operatorname{colim} X_i$ in the category of all algebraic spaces over B.

Proof. We encourage the reader to look instead at Lemma 0GHL and its proof.

Let Z be an algebraic space over B. Suppose that $f_i : X_i \to Z$ is a family of morphisms such that for each $i \to j$ the composition $X_i \to X_j \to Z$ is equal to f_i. We have to construct a morphism of algebraic spaces $f : X \to Z$ over B such that we

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can recover f_i as the composition $X_i \to X \to Z$. Let $W \to Z$ be a surjective étale morphism of a scheme to Z. We may assume that W is a disjoint union of affines and in particular we may assume that $W \to Z$ is separated and that W is separated over B. For each i set $U_i = W \times_{Z,f_i} X_i$ and denote $h_i : U_i \to W$ the projection. Then $U_i \to X_i$ forms an object of $\lim X_{i,spaces,étale}$ with $U_i \to X_i$ separated. By assumption (5) we can find a separated étale morphism $U \to X$ of algebraic spaces and (functorial) isomorphisms $U_i = X_i \times_X U$. By assumption (4) there exists a morphism $h : U \to W$ over B such that the compositions $U_i \to U \to W$ are h_i. Let $g : U \to Z$ be the composition of h with the map $W \to Z$. To finish the proof we have to show that $g : U \to Z$ descends to a morphism $X \to Z$. To do this, consider the morphism $(h, h) : U \times_X U \to W \times_S W$. Composing with $U_i \times_{X_i} U_i \to U \times_X U$ we obtain (h_i, h_i) which factors through $W \times_Z W$. Since $U \times_X U$ is the colimit of the algebraic spaces $U_i \times_{X_i} U_i$ in the category of algebraic spaces separated over B by (4) we see that (h, h) factors through $W \times_Z W$. Hence the two compositions $U \times_X U \to U \to W \to Z$ are equal. Because each $U_i \to X_i$ is surjective and assumption (2) we see that $U \to X$ is surjective. As Z is a sheaf for the étale topology, we conclude that $g : U \to Z$ descends to $f : X \to Z$ as desired. □